Bank Customer Churn Prediction Dashboard

Complete Implementation Guide

Masters in AI for Business (MAIB)
Data Visualization Project

Project Overview:
Complete Docker-based dashboard with ML predictions, real-time data streaming, and 30+ interactive visualizations optimized for classroom projectors.

Technology Stack:
• Docker & Docker Compose
• Python 3.10, FastAPI, Streamlit
• PostgreSQL 15, XGBoost, SHAP
• Plotly Interactive Charts

Data Strategy:
• Kaggle Bank Churn Dataset (10,000 customers)
• Synthetic Data Generation for Live Simulation

Key Deliverables:
• Multi-container Docker architecture
• ML API with SHAP explainability
• 5 dashboard pages with 30+ charts
• Projector-friendly high-contrast design
• Business impact calculator

Create a complete Bank Customer Churn Prediction Dashboard project with Docker, Machine Learning, and live data streaming. The project should use both Kaggle dataset and synthetic data generation with a projector-optimized, visually stunning interface.

PROJECT REQUIREMENTS

1. PROJECTOR-FRIENDLY DESIGN REQUIREMENTS

COLOR SCHEME (High Contrast):

- **Primary Background:** #FFFFFF (Pure White)
- **Secondary Background:** #F8F9FA (Light Gray)
- **Card Background:** #FFFFFF with shadow
- **Text Primary:** #1A1A1A (Almost Black)
- **Text Secondary:** #4A4A4A (Dark Gray)

Risk Colors (High Saturation):

- **High Risk:** #DC3545 (Bright Red)
- **Medium Risk:** #FFA500 (Orange)
- **Low Risk:** #28A745 (Green)

Accent Colors:

- **Primary Blue:** #0066CC (Strong Blue)
- **Success Green:** #28A745
- **Warning Orange:** #FF8C00
- **Danger Red:** #DC3545
- **Info Cyan:** #17A2B8

Chart Colors (Distinct & Bright):

`['#0066CC', '#DC3545', '#28A745', '#FFA500', '#9C27B0', '#00BCD4', '#FF5722', '#4CAF50']`

TYPOGRAPHY (Large & Bold):

- **Page Title:** 48px, Bold, #1A1A1A

- **Section Headers:** 36px, Bold, #2C3E50
- **KPI Values:** 42px, Extra Bold, Color-coded
- **KPI Labels:** 20px, Medium, #4A4A4A
- **Body Text:** 18px, Regular, #1A1A1A
- **Chart Labels:** 16px, Bold, #1A1A1A
- **Chart Values:** 14px, Regular, #1A1A1A

SPACING & SIZING:

- **KPI Cards:** Minimum height 180px
- **Chart minimum height:** 500px
- **Padding around cards:** 24px
- **Margins between sections:** 40px
- **Border radius:** 12px for modern look
- **Box shadows:** Heavy (0 4px 12px rgba(0,0,0,0.15))

DASHBOARD LAYOUT:

- Maximum 3 columns on main views
- KPI cards in a single row at top (5 cards)
- Charts in 2-column grid below KPIs
- Full-width charts for important visualizations
- Sticky sidebar for filters
- Large, obvious buttons (min 44px height)

STREAMLIT CUSTOM CSS:

Create custom CSS that:

- Increases all font sizes by 25%
- Uses high-contrast colors
- Adds drop shadows to all cards
- Makes interactive elements obvious (large buttons, clear hover states)
- Removes Streamlit's default small fonts
- Increases chart legend size

- Makes filters and inputs larger

PLOTLY CHART STYLING:

For all Plotly charts:

- Font size: 16px minimum for all text
- Bold axis titles (20px)
- Large tick labels (14px)
- Thick lines (width: 4)
- Large markers (size: 12)
- High contrast colors
- White background
- Grid lines in light gray (#E0E0E0)
- Legend: 16px, positioned top-right
- Hover labels: 16px, bold values

INTERACTIVE ELEMENTS:

- **Buttons:** Minimum 50px height, 18px font, bold
- **Sliders:** Large handles, clear labels
- **Dropdowns:** 18px font, 50px height
- **Tables:** 16px font, alternating row colors, bold headers
- **Filters:** Large checkboxes, 18px labels

VISIBILITY ENHANCEMENTS:

- All KPI cards have colored left border (8px thick)
- Risk indicators use both color AND icons (■ ■ ■)
- Percentage changes show large arrows (↑ ↓)
- Important insights highlighted in colored boxes
- Charts have thick borders (2px)
- Loading states clearly visible
- Error messages in large, red text

ACCESSIBILITY:

- Minimum contrast ratio: 7:1
- No reliance on color alone (use icons + text)
- Large click targets (minimum 44x44px)
- Clear focus indicators
- Descriptive labels on all elements

2. DOCKER-COMPOSE.YML

Create a docker-compose file with 4 services:

- **PostgreSQL database** (postgres:15-alpine)
- **ML Model API** (FastAPI) on port 8000
- **Streamlit Dashboard** on port 8501
- **Data Simulator** for live streaming

Include health checks, proper networking, and volume mounts.

Services configuration:

```
services:
- database: PostgreSQL with health checks
- ml-api: FastAPI model serving with uvicorn
- dashboard: Streamlit on port 8501
- data-simulator: Python script for live data generation
```

3. ML MODEL API (ml-model/api/model_api.py)

Create a FastAPI application with the following endpoints:

Endpoints:

- **POST /predict** - Single customer churn prediction
- **POST /predict_batch** - Batch predictions
- **GET /model_info** - Model metadata

- **GET /health** - Health check

Features:

- Use XGBoost model
- Include SHAP explanations for predictions
- Return JSON with: customer_id, churn_probability, risk_level (High/Medium/Low), top_features
- Proper error handling and validation
- Type hints using Pydantic models

4. MODEL TRAINING (ml-model/src/train.py)

Create a complete ML pipeline:

- Load data from CSV
- Feature engineering (encode Geography, Gender)
- Train-test split (80-20)
- Train multiple models: Logistic Regression, Random Forest, XGBoost
- Use SMOTE for class imbalance
- Hyperparameter tuning with GridSearchCV
- Save best model, scaler, and feature names as pickle files
- Generate classification report and confusion matrix
- Calculate and save performance metrics

5. DATA SIMULATOR (data/simulate_live.py)

Create a Python script that:

- Generates synthetic bank customers matching Kaggle data distribution
- Uses Faker library for realistic names
- Matches statistical distributions (age, credit score, balance, etc.)
- Inserts new customers into PostgreSQL every 10 seconds

- Calls ML API to get churn predictions
- Logs each insertion with timestamp

Features to match from Kaggle data:

- **CreditScore:** normal distribution (mean ~650, std ~96)
- **Geography:** France (50%), Germany (25%), Spain (25%)
- **Age:** normal distribution (mean ~39, std ~10)
- **Tenure:** 0-10 years uniform
- **Balance:** 0-250000 uniform
- **NumOfProducts:** 1(50%), 2(46%), 3(3%), 4(1%)
- **HasCrCard:** 70% yes, 30% no
- **IsActiveMember:** 52% yes, 48% no
- **EstimatedSalary:** 11-200000 uniform

6. STREAMLIT DASHBOARD (dashboard/app.py)

Create main dashboard page with:

- Page configuration (wide layout, custom title, icon)
- Custom CSS for projector-friendly styling (as specified above)
- Sidebar with filters:
- Geography (multiselect)
- Age Range (slider)
- Product Count (multiselect)
- Active Member status (radio)
- Top KPI cards (5 metrics): Total Customers, Churn Rate, Active Customers, Avg Balance, High Risk Count
- Data loading with @st.cache_data
- Connection to PostgreSQL and ML API
- Auto-refresh every 30 seconds for live data
- Professional footer with branding

7. ADVANCED VISUALIZATIONS (30+ Charts)

Include the following advanced visualizations distributed across 5 dashboard pages:

PAGE 1: OVERVIEW (dashboard/pages/1_■_Overview.py)

Chart 1: Customer Lifetime Value Sunburst

- **Type:** Sunburst Chart (hierarchical)
- **Purpose:** Show customer value segmentation by geography → age group → product count
- **Styling:** Colorful, interactive, circular with large labels

Chart 2: Churn Rate by Geography

- **Type:** Bar chart with gradient colors
- **Purpose:** Compare churn rates across regions
- **Features:** Large text annotations, thick bars, color gradient

Chart 3: Age Distribution with Churn Overlay

- **Type:** Overlapping histograms
- **Purpose:** Show age patterns for churned vs retained customers
- **Features:** Semi-transparent bars, large legend

Chart 4: Customer Journey Sankey Diagram

- **Type:** Sankey Flow Diagram
- **Purpose:** Show customer flow from Geography → Products → Active Status → Churn/Retain
- **Features:** Flowing connections, color-coded paths, thick flows

Chart 5: Temporal Churn Trend

- **Type:** Area Chart with Multiple Layers
- **Purpose:** Show churn trend over time
- **Features:** Stacked by geography or risk level, flowing colored areas

Chart 6: Activity Level Gauge Charts

- **Type:** Multiple Gauge/Speedometer Charts
- **Purpose:** Show overall customer health metrics as speedometers

- **Metrics:** Overall Churn Risk Score, Active Member %, Avg Products per Customer
- **Features:** Red/yellow/green zones, large numbers

PAGE 2: CUSTOMER ANALYSIS (dashboard/pages/2_■_Customer_Analysis.py)

Chart 7: Correlation Heatmap

- **Type:** Annotated Heatmap
- **Purpose:** Show correlations between all features
- **Features:** Color-coded grid with large correlation coefficients

Chart 8: Tenure vs Churn Survival Curve

- **Type:** Kaplan-Meier Survival Curve
- **Purpose:** Show retention rate over customer lifetime
- **Features:** Declining curve with confidence interval shading

Chart 9: Balance vs Salary Quadrant Analysis

- **Type:** Scatter Plot with 4 Quadrants
- **Purpose:** Plot Balance (Y) vs Estimated Salary (X), colored by churn
- **Features:** Clear quadrant lines, large dots, different colors per quadrant

Chart 10: Gender & Age Cohort Heatmap

- **Type:** 2D Heatmap
- **Purpose:** Age groups (rows) × Gender (columns), cell color = churn rate
- **Features:** Large cell annotations, high contrast colors

Chart 11: Balance Brackets Analysis

- **Type:** Stacked Horizontal Bar Chart
- **Purpose:** Divide customers into balance brackets (€0, €1-50K, €50K-100K, etc.)
- **Features:** Large labels, percentage annotations

Chart 12: Interactive Customer Profile Card

- **Type:** Custom styled info card with mini-charts
- **Purpose:** When user selects a customer, show detailed profile
- **Contents:** Risk gauge, feature comparison bars, timeline, recommendations
- **Features:** Professional design, large readable text

Chart 13: Demographic Pyramid

- **Type:** Population Pyramid (back-to-back bar chart)
- **Purpose:** Age distribution, Left = Retained, Right = Churned
- **Features:** Mirror bars, clear age labels

PAGE 3: ML PREDICTIONS (dashboard/pages/3_■_ML_Predictions.py)

Chart 14: Credit Score Segmentation

- **Type:** Stacked Area Chart or Grouped Bar Chart
- **Purpose:** Churn rate for each credit score band across geographies
- **Features:** Credit bands clearly labeled (Poor, Fair, Good, Excellent)

Chart 15: Churn Probability Distribution

- **Type:** Violin Plot or Box Plot
- **Purpose:** Show distribution of churn probabilities across segments
- **Features:** Wide violins, median lines, outlier points

Chart 16: Feature Contribution Waterfall

- **Type:** Waterfall Chart
- **Purpose:** Show how each feature contributes to churn probability
- **Features:** Flowing up/down bars, cumulative total line

Chart 17: Churn Prediction Confidence Intervals

- **Type:** Error Bar Chart or Range Plot
- **Purpose:** Show model's prediction confidence
- **Features:** Large error bars, confidence percentages

Chart 18: 3D Scatter Plot

- **Type:** 3D Scatter Plot (Plotly)
- **Purpose:** Plot customers in 3D space: Age (X), Balance (Y), Credit Score (Z)
- **Features:** Rotatable, color by churn, large markers

Chart 19: Cumulative Churn Curve

- **Type:** Cumulative Line Chart
- **Purpose:** Show cumulative percentage of churned customers by risk score
- **Features:** Steep curve at high risk, annotation of key points

Chart 20: Risk Score Distribution

- **Type:** Donut chart with annotations
- **Purpose:** Show distribution across High/Medium/Low risk categories
- **Features:** Large percentages in center, color-coded segments

Chart 21: Feature Importance Bar Chart

- **Type:** Horizontal bar chart
- **Purpose:** Show top 10 features driving churn predictions
- **Features:** Sorted by importance, percentage values, gradient colors

PAGE 4: MODEL PERFORMANCE (dashboard/pages/4_■_Model_Performance.py)

Chart 22: Monthly Charges Distribution with KDE

- **Type:** Histogram with KDE overlay
- **Purpose:** Show distribution of account balances for churned vs retained
- **Features:** Two overlapping distributions, smooth KDE line

Chart 23: Product Mix Matrix

- **Type:** Heatmap Grid
- **Purpose:** Has Credit Card (rows) × Is Active Member (cols), cell = churn rate
- **Features:** Large cell values, count annotations

Chart 24: Model Accuracy Over Time

- **Type:** Line chart with confidence band
- **Purpose:** Show how model accuracy improves with more data
- **Features:** Accuracy line trending up, shaded confidence interval

Chart 25: ROC Curve

- **Type:** Line plot with AUC score
- **Purpose:** Show model's true positive vs false positive rate
- **Features:** Diagonal reference line, AUC score prominently displayed

Chart 26: Confusion Matrix Heatmap

- **Type:** 2x2 Heatmap with annotations
- **Purpose:** Show True Positives, False Positives, True Negatives, False Negatives
- **Features:** Large numbers, color gradient, percentages

Chart 27: Precision-Recall Curve

- **Type:** Line plot
- **Purpose:** Show precision vs recall trade-off
- **Features:** Optimal point marked, clear labels

Chart 28: Model Comparison Table

- **Type:** Styled DataFrame/Table
- **Purpose:** Compare Logistic Regression, Random Forest, XGBoost performance
- **Features:** Color-coded best values, sortable columns

PAGE 5: BUSINESS IMPACT (dashboard/pages/5_■_Business_Impact.py)

Chart 29: Impact Simulator (Interactive)

- **Type:** Interactive slider-controlled bar chart
- **Purpose:** User adjusts retention campaign parameters
- **Features:** Real-time updates, ROI calculation, large result display

Chart 30: Engagement Score Components

- **Type:** Stacked bar with formula display
- **Purpose:** Show how Customer Health Score is calculated
- **Features:** Component breakdown, 0-100 scale with color coding

Chart 31: Business Impact Dashboard

- **Type:** Mixed - KPI cards + charts
- **Purpose:** Translate ML predictions into business metrics
- **Metrics:** Revenue at risk, Cost of acquisition vs retention, CLV by segment
- **Features:** Large currency values, trend indicators

Chart 32: Customer Count Funnel

- **Type:** Funnel Chart
- **Purpose:** Show customer count at each engagement level
- **Features:** Narrowing funnel, drop-off percentages

Chart 33: Risk Score Meter per Geography

- **Type:** Grouped Gauge Charts
- **Purpose:** One speedometer per geography showing average risk

- **Features:** Three large gauges side-by-side, color zones

Chart 34: Product Adoption Matrix Bubble Chart

- **Type:** Bubble Chart
- **Purpose:** X=Products, Y=Tenure, Size=Count, Color=Churn Rate
- **Features:** Varying bubble sizes, interactive tooltips

ADDITIONAL ADVANCED CHARTS (Optional but Impressive):

Chart 35: Customer Segmentation Radar Chart

- **Type:** Radar/Spider Chart
- **Purpose:** Compare different customer segments across multiple dimensions
- **Features:** Overlapping polygons, 6-8 dimensions

Chart 36: Network Graph of Customer Characteristics

- **Type:** Network/Node Graph
- **Purpose:** Show relationships between customer attributes
- **Features:** Circular network, edge thickness = correlation strength

8. COMPONENT FILES

dashboard/components/kpi_cards.py

Create reusable KPI card component:

- Large value display (52px)
- Icon support
- Delta with arrows
- Color-coded borders
- Hover effects

dashboard/components/charts.py

Create standard chart functions:

- Consistent styling across all charts
- PLOTLY_LAYOUT template applied
- High contrast colors
- Large fonts
- Thick lines and markers

dashboard/components/advanced_charts.py

Create advanced visualization functions:

- Sunburst charts
- Sankey diagrams
- 3D scatter plots
- Gauge charts
- Network graphs
- Waterfall charts

9. README.md

Create comprehensive README with:

- Project overview and business problem
- Architecture diagram (ASCII art)
- Prerequisites (Docker, Docker Compose)
- Installation steps
- How to download Kaggle dataset
- How to run the project
- How to access services (URLs and ports)
- Dataset information and features
- Technology stack
- Features list with screenshots placeholders
- ML model details and performance
- Troubleshooting section
- Future enhancements
- Contributors and license

10. ADDITIONAL FEATURES TO IMPLEMENT

Auto-Refresh Mechanism

- Dashboard refreshes every 30 seconds
- Manual refresh button
- Display last update timestamp

Dark Mode Toggle

- Optional dark theme
- User preference saved in session state

Data Quality Indicators

- Show completeness percentage
- Flag anomalies
- Data freshness indicator

Tooltips and Help Text

- Hover tooltips on all metrics
- Info icons with explanations
- Onboarding tour for first-time users

Loading States

- Skeleton loaders for charts
- Progress bars for data loading
- Smooth transitions

Error Handling

- Graceful error messages
- Retry mechanisms
- Fallback data display